# AI-COMPASS: A FRAMEWORK FOR IDENTIFYING HIGH-IMPACT AI USE CASES

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**ABSTRACT**

Despite the rapid proliferation of artificial intelligence (AI) across industries, a significant proportion of AI projects fail to deliver meaningful return on investment (ROI). This paper introduces AI-Compass, a structured framework for identifying high-potential AI use cases that align with business goals, technical feasibility, and economic viability. Building upon the well-established Impact-Competition-Effort (ICE) model, AI-Compass incorporates additional dimensions critical to AI deployment, including model maturity, cost of error, and data availability. This research outlines the fundamental principles of AI-Compass, presents case studies to validate its effectiveness, and provides insights into optimizing AI investments for enterprises.

***Keywords*** AI System Development · ROI in traditional software · Phases of AI Development

## 1 Introduction

The failure rate of AI initiatives remains high due to challenges in selecting optimal use cases. Traditional frameworks such as the ICE model are inadequate for AI-specific considerations, as they fail to account for the stochastic nature of AI models, high implementation costs, and data dependencies. This study presents AI-Compass, a refined framework that systematically evaluates AI use cases based on multiple industry-relevant factors.

## 2 Limitations of Traditional Use Case Selection Models

The ICE framework—Impact vs. Competition vs. Effort—has been widely used in software product development. However, AI-specific use case selection requires additional considerations:

1. AI's Stochastic Nature: Unlike deterministic software, AI models produce probabilistic outputs, necessitating mechanisms like human-in-the-loop systems.
2. High Development and Deployment Costs: Effective AI solutions demand substantial computational resources, specialized talent, and prolonged iteration cycles.
3. Integration Complexity: AI adoption extends beyond system integration; it requires process changes, metric instrumentation, and operational orchestration.

These factors necessitate a new framework tailored for AI investments.

## 3 The AI-Compass Framework

AI-Compass extends the ICE framework by incorporating five additional parameters:

- Traditional Engineering Shortfalls: AI should be considered only when conventional engineering solutions prove inadequate.
- Model Maturity: Viability depends on the existence of sufficiently developed algorithms that align with the problem statement.
- Cost of Error: AI should be deployed in scenarios where occasional mistakes are acceptable without catastrophic consequences.
- Economic Viability: The cost-benefit ratio must favor AI implementation, with a preference for revenue-generating use cases over cost-cutting initiatives.
- Data Availability: High-quality, domain-specific data is essential for AI model performance and scalability.

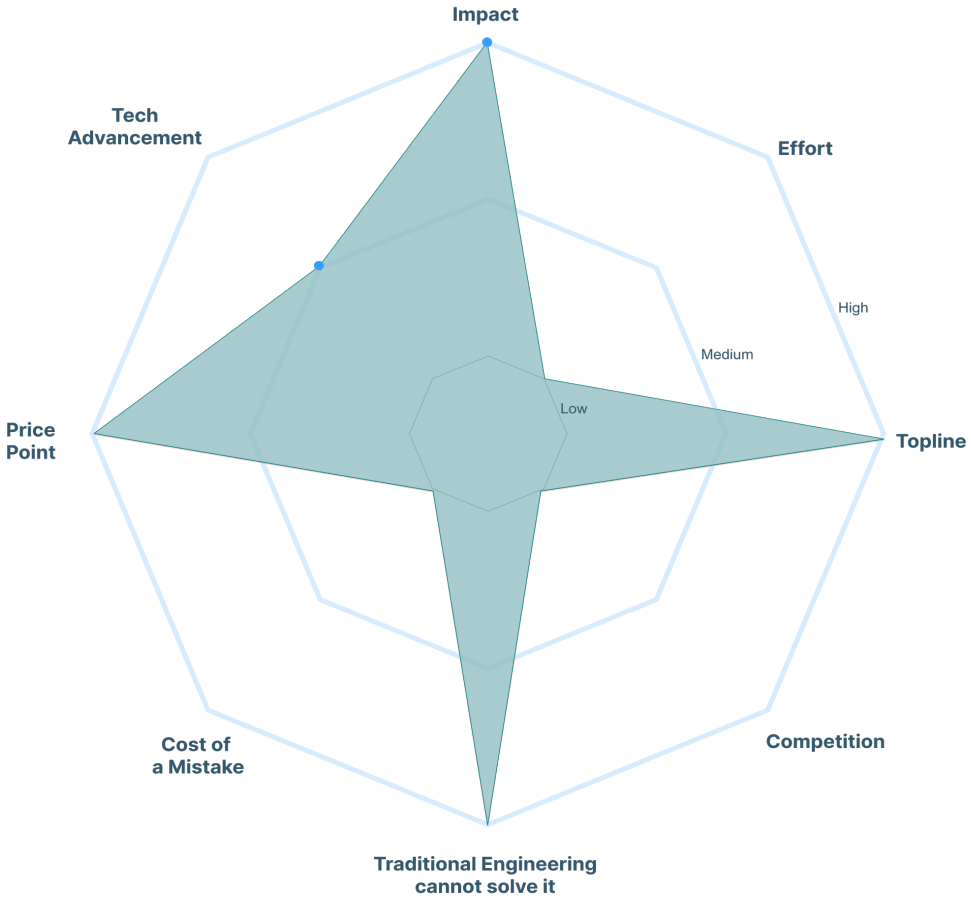


Figure 1: AI-Compass framework

## 4 Case Studies

To demonstrate AI-Compass in practice, we analyze two real-world use cases.

### 4.1 Special Character Recognition in Google Docs

- Problem Statement: Users struggle to find special characters in Google Docs, requiring an AI-driven solution for quick retrieval.
- Solution: Google employed Sketch-RNN, a stroke-based neural network, allowing users to draw symbols for recognition.
- AI-Compass Evaluation:

  - High Impact: Enhances user experience and efficiency.
  - Low Effort: Leveraged an existing AI model.
  - Low Cost of Mistake: Incorrect predictions result in minor inconvenience.
  - Economic Viability: Feature enhancement supports customer retention without major cost overhead.
  - Data Availability: Google's QuickDraw dataset provided extensive training data.

### 4.2 AI Chatbots for Customer Support

- Problem Statement: Automating customer service to reduce operational costs.
- Solution: AI-powered chatbots attempt to replace human agents.
- AI-Compass Evaluation:
  - High Competition: Market is saturated with chatbot solutions.
  - High Effort: Requires continuous model fine-tuning and human intervention.
  - High Cost of Mistake: Incorrect responses degrade customer trust and satisfaction.
  - Poor Economic Viability: Cost savings from automation often do not offset AI maintenance expenses.
  - Data Limitations: High variability in customer queries reduces model accuracy.

This analysis indicates that AI chatbots, while appealing in theory, often fail to meet economic and technical feasibility criteria.

## 5 Conclusion

AI investments must be guided by a structured framework that evaluates feasibility across multiple dimensions. AI-Compass provides a comprehensive methodology for selecting viable AI use cases, balancing impact, effort, economic viability, and technical constraints. Organizations adopting AI-Compass can maximize ROI while minimizing risk. Future research will focus on refining the framework through additional industry applications and empirical validation.

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